

CONTINUOUS TIME KALMAN BUCY FILTER FOR GPS WITH NON LINEAR EXTENSION

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Compiled on March 17, 2026

1 Introduction

This project studies state estimation for a nonlinear dynamical system arising in satellite-based positioning. The receiver state is inferred from noisy pseudorange measurements obtained from multiple satellites. Since the observation model is nonlinear in the state, classical linear filtering techniques cannot be applied directly.

To address this, we consider two approaches based on linearization: the Linearized Kalman–Bucy Filter (KBF), where the system is linearized about a nominal trajectory, and the Extended Kalman–Bucy Filter (EKBF), where linearization is performed about the current estimate. The performance of these filters is evaluated under different satellite geometries and initial conditions, highlighting the impact of nonlinearity and model mismatch on estimation accuracy.

The implementation corresponding to this project is available at: [GitHub Repository](#).

2 Theoretical Analysis

2.1 Dynamic and Observation Models

Let the state be $\mathbf{x}_t \in \mathbb{R}^d$ with dynamics

$$d\mathbf{x}_t = f(\mathbf{x}_t, t) dt + G_t d\mathbf{w}_t, \quad (1)$$

where $\mathbf{w}_t$ is a standard Wiener process.

The measurement model is nonlinear:

$$d\mathbf{y}_t = h(\mathbf{x}_t)dt + \mathbf{R}^{\frac{1}{2}}d\mathbf{V}_t, \quad (2)$$

where $\mathbf{v}_t \sim \mathcal{N}(0, R)$.

Since $h(\cdot)$ is nonlinear, we linearize it as

$$h(\mathbf{x}_t) \approx h(\bar{\mathbf{x}}_t) + H_t(\mathbf{x}_t - \bar{\mathbf{x}}_t), \quad (3)$$

where

$$H_t = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\bar{\mathbf{x}}_t}. \quad (4)$$

2.2 Kalman–Bucy Filter Equations

1. Linearization about a nominal trajectory – Linearized-KBF

Let $\bar{\mathbf{x}}_t$ be a nominal trajectory. The system is linearized about $\bar{\mathbf{x}}_t$.

2.3 State Estimate

$$d\hat{\mathbf{x}}_t = f(\bar{\mathbf{x}}_t, t) dt + K_t (d\mathbf{y}_t - h(\bar{\mathbf{x}}_t) dt - H_t(\hat{\mathbf{x}}_t - \bar{\mathbf{x}}_t) dt) \quad (5)$$

2.4 Covariance Update

$$\frac{dS_t}{dt} = A_t S_t + S_t A_t^\top + Q - S_t H_t^\top R^{-1} H_t S_t \quad (6)$$

2.5 Kalman Gain

$$K_t = S_t H_t^\top R^{-1} \quad (7)$$

The nominal trajectory is precomputed by simulating the system dynamics discounting the process noise.

2. Linearization about the previous estimate – EKBF

In the EKBF, linearization is performed about the current estimate $\hat{\mathbf{x}}_t$.

2.6 State Estimate

$$d\hat{\mathbf{x}}_t = f(\hat{\mathbf{x}}_t, t) dt + K_t (d\mathbf{y}_t - h(\hat{\mathbf{x}}_t) dt) \quad (8)$$

2.7 Covariance Update

$$\frac{dS_t}{dt} = A_t S_t + S_t A_t^\top + Q - S_t H_t^\top R^{-1} H_t S_t \quad (9)$$

where

$$H_t = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_t} \quad (10)$$

2.8 Kalman Gain

$$K_t = S_t H_t^\top R^{-1} \quad (11)$$

3 Experiments and Analysis

This section summarizes the experimental results and discusses their implications.

3.1 Repository Description

The repository is organized into the folders `src`, `results`, and a set of experiment scripts (including `demo.py`). The `src` folder contains the core implementation for simulating the system dynamics. The script `demo.py` is used to generate the data, which is subsequently processed by the experiment scripts to produce the figures.

3.2 Experiment 1: Baseline Run (Well Spread Satellites)

Implementing the Extended Kalman–Bucy Filter (EKF) under a well-spread satellite geometry yields the following results. The estimated trajectory closely tracks the true trajectory, indicating stable and accurate state estimation despite the presence of initial uncertainty. This is further reflected in the evolution of the covariance ellipses, which progressively shrink over time, demonstrating the reduction in estimation uncertainty as more measurements are assimilated. These ellipses provide a geometric interpretation of the state covariance, capturing both the magnitude and directional structure of uncertainty.

Note that RMSE has been computed for a single experiment and not by Monte Carlo approximation.

The baseline innovation norm remains bounded between 0 and 2 m, indicating that the measurement residuals are consistent with the model predictions.

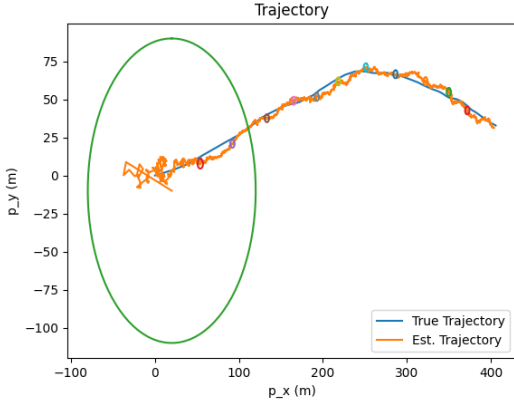


Figure 1: True vs estimated 2D trajectory

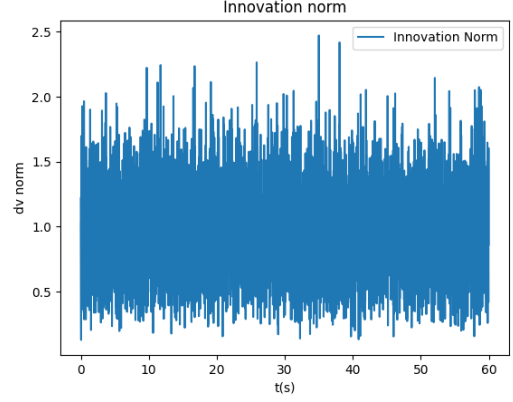


Figure 2: Innovation process norms

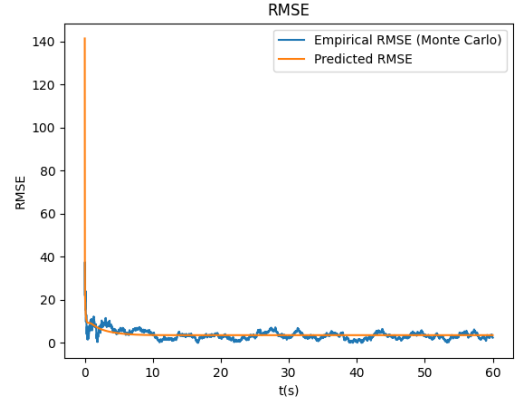


Figure 3: RMSE variation

3.3 Experiment 2: Degenerate geometry study

Poor satellite placement (clustering) leads to the observation Jacobian having nearly dependent rows, resulting in a loss of information. In this case, the clustering occurs along the x-direction, which is reflected in the covariance ellipses exhibiting significantly higher uncertainty in that direction. Compared to the baseline geometry, the ellipses are noticeably larger, indicating degraded estimation accuracy. The experiment also shows a higher innovation norm and poor convergence of the RMSE error, further highlighting the impact of poor geometry on filter performance.

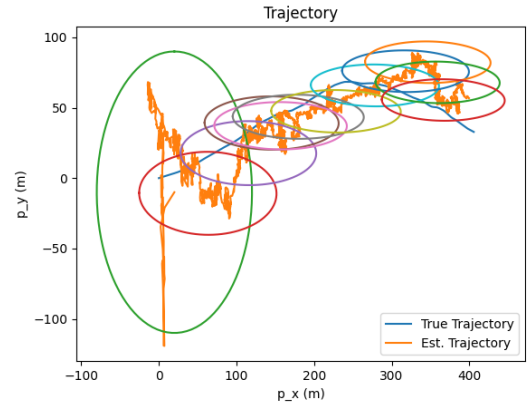


Figure 4: True vs estimated 2D trajectory

Comparison of speed with baseline

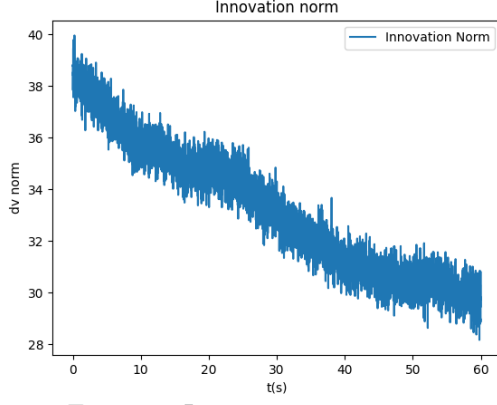


Figure 5: Innovation process norms

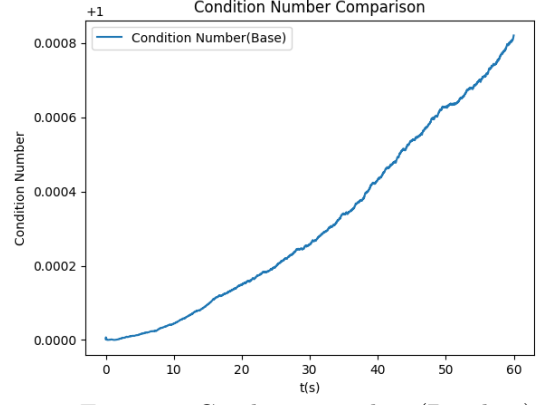


Figure 8: Condition number (Baseline)

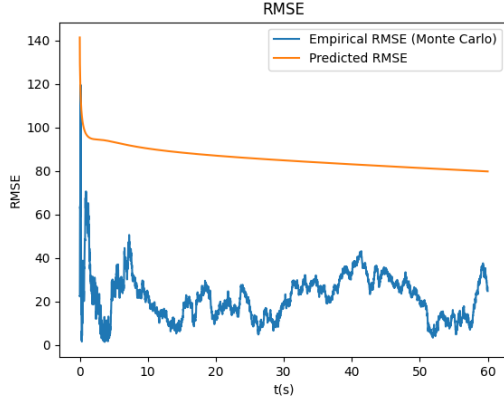


Figure 6: RMSE variation

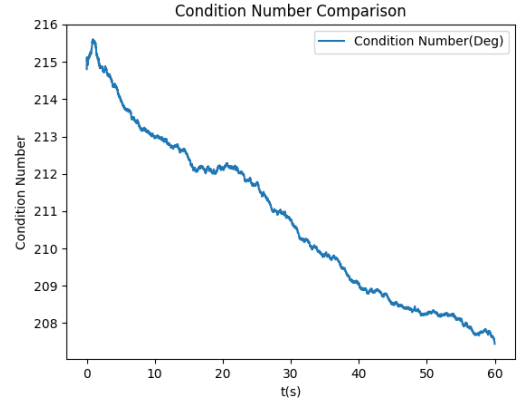


Figure 9: Condition number (Degenerate)

The RMSE error has a better convergence for the baseline geometry as compared to the degenerate case.

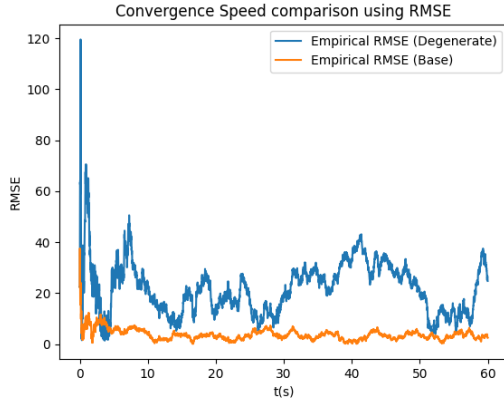


Figure 7: RMSE Comparison

Condition Number variation

The condition number for the degenerate geometry is orders of magnitude higher since the Jacobian is nearly non singular.

3.4 Experiment 3: Initialization sensitivity

The initial error is set to (300, 400) m. We observe that even under large initial error, the EKBF converges to the true trajectory. This performance is further supported by the shrinking covariance ellipses, bounded innovation norms, and convergence of the empirical RMSE error.

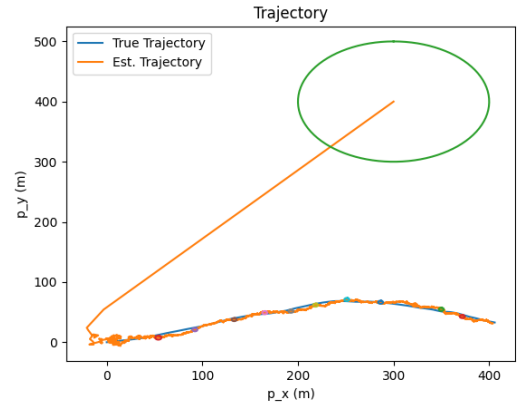


Figure 10: EKBF trajectory with large initial error

The KBF performs considerably worse than the EKBF under large initial error, as the linearization is performed about a nominal trajectory that is significantly offset from the true state due to the large initial error.

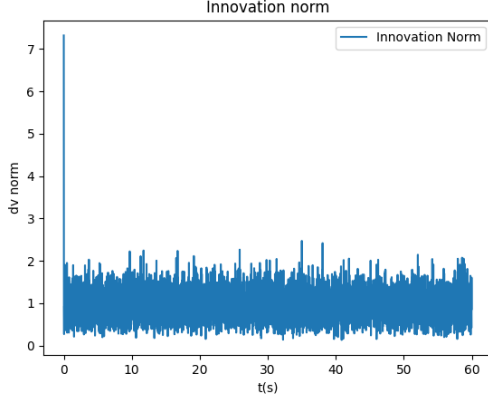


Figure 11: EKF innovation with large initial error

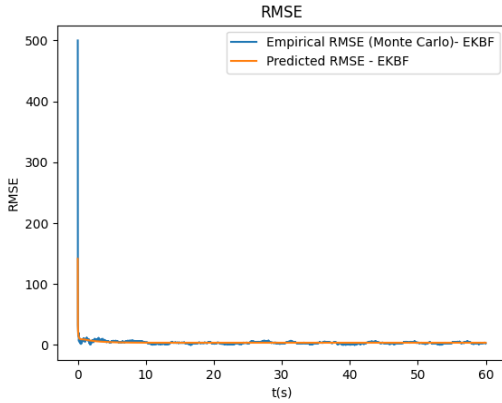


Figure 12: EKF RMSE with large initial error

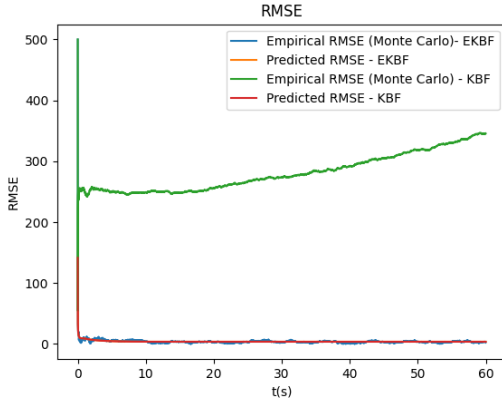


Figure 13: EKF vs KBF for large initial error

4 Discussion

1. Why are at least 3 satellites required in 2D when clock bias is unknown?

In the ideal case with no clock bias, each satellite provides a distance measurement, which geometrically corresponds to a circle centered at the satellite. With two satellites, the receiver lies at one of the intersection points of the two circles (typically two possible solutions). When the clock bias is unknown, each pseudorange measurement includes an additional unknown offset. Thus, in 2D, the unknowns are (x, y) and the clock bias b , giving three unknowns. Each satellite provides one equation,

so at least three satellites are required to obtain a determined system. Geometrically, the unknown clock bias shifts the radii of all circles by the same amount, so two measurements are insufficient to localize the receiver uniquely. A third satellite resolves this ambiguity. Similarly, in 3D, four satellites are required to estimate (x, y, z) and the clock bias.

2. How does satellite geometry affect the term $S_t H_t^T R^{-1} H_t S_t$ and the Riccati evolution? Relate to GDOP.

As discussed previously, H_t contains the direction vectors to the satellites. When the geometry is ill-conditioned, the matrix $H_t^T R^{-1} H_t$ has eigenvalues of small magnitude. These eigenvalues can be interpreted as the amount of information along corresponding directions. Hence, small eigenvalues indicate a lack of information in those directions, leading to a reduced rate of uncertainty reduction.

3. Under what conditions does the EKF diverge?

The EKF relies on local linearization of the nonlinear dynamics about the current estimate. This assumption can break down when the estimate is significantly erroneous, such as in the presence of large initial errors, leading to inaccurate linearization and potential divergence. The filter may also fail when the system dynamics or observation model are highly nonlinear, where a first-order approximation is insufficient to capture the true behavior.

Additionally, discretization and numerical issues can contribute to instability; for example, if the time step is too large, the linearized model may no longer accurately represent the continuous dynamics. Additionally, as we have seen in the experiments, the geometry might also affect the convergence of the filter.

4. Does the innovation process behave like Brownian motion?

Under the ideal assumptions of the Kalman–Bucy filter (linear system, Gaussian noise, correct model), the innovation process is a white Gaussian process, with independent increments. More precisely, it behaves like a Brownian motion after appropriate normalization (i.e., a Wiener process when scaled by the innovation covariance). However, when these assumptions break down—such as in the presence of nonlinear dynamics, model mismatch, or incorrect noise statistics—the innovation process is no longer white. The increments become correlated and may exhibit bias, so it no longer behaves like Brownian motion.